BTS–VI–04.18–0158 Reg. No.

C

# B.Tech. Degree VI Semester Examination April 2018

## ME 602 DYNAMICS OF MACHINERY

*(2006 Scheme)*

Time: 3 Hours Maximum Marks: 100

### PART A

(Answer *ALL* questions)

($8 \times 5 = 40$)

I. (a) What is meant by dynamically equivalent system? Explain.

(b) Derive an expression for the torque exerted on the crankshaft when friction and inertia of moving parts are neglected.

(c) Define the terms 'coefficient of fluctuation of energy' and 'coefficient of fluctuation of speed'.

(d) Derive the necessary condition of stability for a centrifugal governor.

(e) What are the advantages of coupled locomotives over uncoupled locomotives?

(f) Explain the working principle of balancing machines.

(g) Explain the working of (i) rope brake dynamometer and (ii) belt transmission dynamometer with neat figures.

(h) Derive an expression for the braking torque of an internal expanding shoe brake.



### PART B

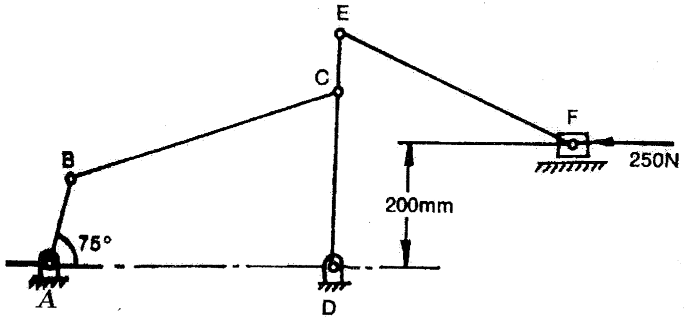
($4 \times 15 = 60$)

II. The connecting rod of a vertical high speed engine is 600 mm long between centres and has a mass of 3 kg. Its centre of mass lies at 200 mm from the big end bearing. When suspended as a pendulum from the gudgeon pin axis, it makes 45 complete oscillations in 30 seconds. The piston stroke is 250 mm. The mass of the reciprocating parts is 1.2 kg. Determine the Inertia torque on the crankshaft when the crank makes an angle of 140° with top dead centre. The engine speed is 1500 rpm.

**OR**

III. For the static equilibrium of the mechanism shown in figure, find the required input torque on link AB.

The dimensions are: AB = 150 mm, BC = AD = 500 mm, DC = 300 mm, CE = 100 mm and EF = 450 mm.




(P.T.O.)

IV. A punching press is required to punch 400 mm diameter holes in a plate of 30 mm thickness at the rate of 4 holes per minute. It requires 6 Nm of energy per $mm^2$ of sheared area. The punch has a stroke of 100 mm. The speed of the flywheel varies from 320 rpm to 280 rpm. If the radius of gyration of flywheel is 1 m, find the mass of the flywheel.

**OR**

V. A ship has a propeller of mass moment of inertia of 2000 $kg.m^2$. The propeller rotates at a speed of 360 rpm in the clockwise sense looking from stern. Determine (i) gyroscopic couple and its effect when ship moves at 30 km/hr, and steers to the left at a radius of 200 m. (ii) Maximum gyroscopic couple and its effect when ship pitches up with an amplitude of 10° and time period of 20 seconds. The motion occurs with SHM.

VI. A 270 cm long shaft carries three pulleys, two at its ends and the third at midpoint. The two end pulleys have masses 48 kg and 20 kg and their centre of gravities are 1.5 cm and 1.25 cm respectively from the axis of the shaft. The middle pulley has 56 kg mass and its centre of gravity is 1.5 cm from the shaft axis. The pulleys are so keyed to the shaft that the assembly is in static balance. The shaft rotates at 300 rpm in two bearings 180 cm apart with equal overhungs on either side. Determine the relative angular position of the pulleys.

**OR**

VII. The two cylinders of a 60° V engine are placed symmetrically. The two connecting rods are coupled directly to a single crank. The stroke is 100 mm and length of each connecting rod is 165 mm. The mass of the reciprocating parts per cylinder is 1.5 kg. Determine the values of primary force when crank is rotating at a speed of 2000 rpm clockwise.

VIII. Determine the maximum power that can be transmitted using a belt of 100 mm × 10 mm with an angle of lap of 160°. The density of belt is 0.001 $gm/mm^3$ and the coefficient of friction may be taken as 0.25. The tension in the belt must not exceed 1.5 $N/mm^2$

**OR**

IX. The external and internal radii of a friction plate of a single clutch are 120 mm and 60 mm respectively. The total axial thrust with which the friction surfaces are held together is 1500 N. For uniform wear, find the maximum, minimum and average pressure on the contact surfaces.

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